Adapting Foundation Models for Astronomical Time Series Classification: Setting 1 on the LSST/PLAsTiCC Benchmark

Zié Coulibaly [1] **Clijo Jose** [1] **Oumouhani El Vilaly** [1]

Abstract

We address 14-class astronomical object classification on the LSST/PLAsTiCC benchmark (The PLAsTiCC Team, 2018) a multivariate time series dataset of shape $(N, T{=}36, C{=}6)$ with severe class imbalance (155:1 ratio) and sparse observations. Following Setting 1, we adapt two pre-trained foundation models: MOMENT-1-large (Goswami et al., 2024) and Chronos-T5-Small (Ansari et al., 2024). We further compare with UniTS (Gao et al., 2024), a non-pretrained dual-attention architecture trained with data augmentation. Our best ensemble combining Chronos, InceptionTime×5, PatchTST, and MultiROCKET via F1-weighted soft voting achieves **64.48% accuracy** and **Weighted F1 = 0.61**, comparing favourably to the best published result on this benchmark (MUSE, 63.6%, Ruiz et al. 2021). Code: `https://github.com/coulzie031/DeepLearningForTimeSeries`

1. Introduction

The LSST (Large Synoptic Survey Telescope) dataset, derived from the PLAsTiCC challenge, is one of the most challenging multivariate time series classification (TSC) benchmarks. Each light curve records the brightness of an astronomical object across 36 epochs and 6 photometric filters (passbands u, g, r, i, z, y), yielding a tensor of shape $(N, 36, 6)$. Two factors make this dataset particularly hard: (i) *sparse observations* leading to NaN values, and (ii) *extreme class imbalance* class 53 (kilonovae) has only 7 training samples out of 2,459 total.

Foundation models pre-trained on large time-series corpora offer a natural approach to scarce-data regimes. We explore two such models under *Setting 1*: adapt a pre-trained foundation model to the classification task. We further investigate UniTS (Gao et al., 2024), a from-scratch dual-attention architecture combined with targeted data augmentation, as a strong competitor. Finally, we combine complementary models through an ensemble to push accuracy beyond the published SOTA.

2. Dataset and Preprocessing

LSST/PLAsTiCC. The dataset contains 2,459 training samples, 2,466 test samples, and 14 classes (supernovae subtypes, variable stars, AGN, kilonovae, microlensing events). Labels are heavily imbalanced: the dominant class (SN Ia, class 90) represents 44% of training data while class 53 has 7 samples.

Preprocessing. Missing values are filled via forward-fill, backward-fill, then zero-fill. A binary mask (B, T, C) encodes observation positions. All channels are z-normalized using training statistics. To address imbalance, we use a `WeightedRandomSampler` with inverse-frequency weights and Focal Loss ($\gamma = 2$) (Lin et al., 2017), which downweights easy majority examples.

Data augmentation (applied only during training): *Jitter* adds Gaussian noise ($\sigma{=}0.05$) with 50% probability; *scaling* multiplies each channel by a random factor in $[0.9, 1.1]$; *time shift* applies a circular roll of ± 3 steps with 30% probability; *MixUp* ($\alpha{=}0.3$) interpolates pairs of samples and their labels.

3. Methods

3.1. Baseline: InceptionTime

InceptionTime (Ismail Fawaz et al., 2020) applies multiscale 1D convolutions to each channel independently. We use kernel sizes $(9, 19, 39)$, 2 residual Inception blocks, $F{=}32$ filters, and Global Average Pooling totalling 474k parameters. No pre-training is used. Trained with Focal Loss, AdamW, and cosine annealing, early-stopped on validation macro F1.

[1]Deep Learning for Time Series, 2026.**AUTHORERR: Missing \icmlcorrespondingauthor.**

Proceedings of the 43^{rd} International Conference on Machine Learning, Seoul, South Korea. PMLR 306, 2026.

3.2. Setting 1 Foundation Model Adaptation

MOMENT-1-large (Goswami et al., 2024). A 341M-parameter T5-based encoder pre-trained on over 1B timesteps from diverse domains. We pad sequences to T=40 (multiple of patch size 8), extract per-channel embeddings of dimension 1024, and flatten to a 6144-dim vector fed to an MLP head. Training proceeds in two phases: (1) linear probing backbone frozen, only the head trains (200 epochs, early stop ep62, val F1 = 0.303); (2) partial unfreeze last 4 of 24 T5 encoder blocks unfrozen with LR=5×10^{-6} (100 epochs, early stop ep26, val F1 = 0.303). Despite native support for classification, MOMENT fails to improve: 341M parameters is an ill-matched scale for 1,967 training samples.

Chronos-T5-Small (Ansari et al., 2024). A 46M-parameter T5-based model originally pre-trained for univariate probabilistic *forecasting* on real and synthetic time series. We discard the decoder and repurpose the encoder for classification analogous to using BERT for sentence classification. Each of the 6 passbands is tokenized independently; masked mean-pooling over hidden states yields a 512-dim per-channel embedding; the 6 embeddings are concatenated into a 3072-dim representation, which is fed to a 3-layer MLP head. Training: phase 1 linear probing (200 ep, stop ep86, val F1 = 0.433); phase 2 partial unfreeze of last 4 encoder blocks, $\mathrm{LR}_{\mathrm{head}}=5\times10^{-5}$, $\mathrm{LR}_{\mathrm{enc}}=5\times10^{-6}$ (100 ep, stop ep32, val F1 = 0.433). Chronos outperforms MOMENT decisively despite being designed for forecasting, confirming that model scale matters more than task alignment when data is scarce.

3.3. Competitor: UniTS with Data Augmentation

UniTS (Gao et al., 2024) is a NeurIPS 2024 model designed for unified multi-task time series learning. Its architecture interleaves three operations per block: (i) *sequence attention* over the time axis; (ii) *variable attention* across channels; (iii) a *Dynamic Linear Operator* (DLO) that performs low-rank temporal mixing via a factorized weight matrix $W=UV^\top$, $U\in\mathbb{R}^{T\times r}$, $V\in\mathbb{R}^{T\times r}$. A learnable `[TASK]` token is prepended to the sequence. We use d=128, 8 heads, 6 blocks, DLO rank r=16 ($\sim$1.6M params). Trained with label smoothing (ε=0.1), AdamW, cosine annealing, and the three augmentations described above. UniTS achieves the highest single-model accuracy (66.06%) and macro F1 (0.5314).

3.4. Additional Models

InceptionTime-Large ×5. Five independently seeded instances of a larger InceptionTime (F=64, 3 blocks, 1.8M params), trained with distinct data splits and augmentation seeds. Test-time augmentation (TTA, N=5 passes) is applied. The averaged probabilities yield acc = 0.6456.

PatchTST (Nie et al., 2023). Channel-independent Transformer: each channel is split into non-overlapping patches of length 4 (9 patches for T=36), linearly projected to d=64, and processed by 3 Transformer encoder layers with 4 heads. Classification head pools over patch representations. Trained with MixUp and TTA. Achieves acc = 0.5985, macro F1 = 0.4599.

MultiROCKET (Dempster et al., 2020). 79,968 random convolutional kernel features followed by a RidgeClassifierCV. Non-neural, no augmentation. Achieves acc = 0.6079.

3.5. Ensemble: F1-Weighted Soft Voting

Each neural model outputs a probability vector $\mathbf{p}_i \in \mathbb{R}^{14}$ over classes. We combine models via weighted soft voting:

$$\hat{y} = \arg\max_c \sum_i w_i\, p_{i,c}, \quad w_i = \frac{\exp(10 \cdot F1_{\mathrm{val},i})}{\sum_j \exp(10 \cdot F1_{\mathrm{val},j})} \tag{1}$$

Models with validation macro F1 $<$ 0.33 are excluded. The final ensemble includes InceptionTime-Large×5 (w=0.852), PatchTST (w=0.068), Chronos (w=0.051), and MultiROCKET (w=0.029). MOMENT is excluded (val F1 = 0.303).

4. Results

Table 1 summarises all results on the LSST test set (2,466 samples). Macro F1 is low due to two structurally unlearnable classes: class 53 (7 train samples, F1 = 0) and class 64 (24 samples, F1 = 0.08). Excluding these two classes, macro F1 rises to $\approx$0.49. Weighted F1 = 0.61 (weighted by class support) better reflects real-world performance on the learnable 90% of test samples.

Foundation model comparison. Chronos (46M) significantly outperforms MOMENT (341M): +22.5 pts accuracy, +15.4 pts macro F1. The key factor is scale: at 1,967 training samples, a 341M-parameter model is prone to underfitting even under linear probing, while Chronos' compact encoder fine-tunes effectively. Chronos also surpasses the baseline (macro F1 0.433 vs. 0.375), validating Setting 1.

Competitor comparison. UniTS[†] achieves the highest single-model scores among all tested methods (acc = 0.6606, F1 = 0.5314), driven by its dual sequence–variable attention and low-rank temporal mixing. Our ensemble reaches acc = 0.6448, above the best published result on this benchmark (MUSE, 63.6%, Ruiz et al. 2021), though a direct controlled comparison is limited since Ruiz et al. (2021) reports accuracy only and does not address class imbalance.

Table 1. Test set results on LSST (14-class). Acc = accuracy, Mac.F1 = macro F1, W.F1 = weighted F1. †External implementation (github.com/mhani6/deep_learning_time_series). SOTA from Ruiz et al. (2021) reports accuracy only.

Method	Acc	Mac.F1	W.F1
Baselines (no pre-training)			
InceptionTime	0.5483	0.3753	
InceptionTime + aug.†	0.5783	0.4324	
Foundation models (Setting 1)			
MOMENT-1-large	0.3078	0.2791	
Chronos-T5-Small	0.5333	0.4333	
Competitor models			
PatchTST + aug.† + TTA	0.5985	0.4599	
MultiROCKET	0.6079	0.3625	
UniTS + aug.†	0.6606	**0.5314**	
InceptionTime-L×5 + TTA	0.6427	0.4150	
Ensemble (ours)	**0.6448**	0.4200	**0.61**
Published SOTA (Ruiz et al., 2021)			
MUSE	0.636		
ROCKET	0.632		
MrSEQL	0.603		

Class imbalance. Both approaches substantially benefit from imbalance handling. Standard InceptionTime from the bake-off (no imbalance handling) achieves only 34.0% (Ruiz et al., 2021); our baseline with WeightedSampler and Focal Loss reaches 54.8% a +20.8 pt gain from preprocessing alone.

5. Conclusion

We present a comprehensive study of foundation model adaptation (Setting 1) and complementary architectures on the LSST benchmark. Our key findings are: (1) Chronos-T5-Small, despite being designed for forecasting, transfers effectively to multivariate classification, outperforming MOMENT by a large margin confirming that model scale relative to training set size matters more than pre-training task alignment; (2) UniTS with data augmentation achieves the best single-model macro F1 (0.5314), showing the complementary value of dual-attention architectures; (3) our ensemble (64.48%) compares favourably to the best published result on this benchmark (Ruiz et al., 2021), without any additional data; (4) imbalance handling (WeightedSampler + Focal Loss) is the single most impactful preprocessing decision on this dataset.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Arango, S. P., Kapoor, S., et al. Chronos: Learning the language of time series. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2403.07815.

Dempster, A., Petitjean, F., and Webb, G. I. ROCKET: Exceptionally fast and accurate time series classification using random convolutional kernels. *Data Mining and Knowledge Discovery (DMKD)*, 34:1454–1495, 2020.

Gao, S., Koker, T., Queen, O., Hartvigsen, T., Tsiligkaridis, T., and Zitnik, M. UniTS: A unified multi-task time series model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2403.00131.

Goswami, M., Szafer, K., Choudhry, A., Cai, Y., Li, S., and Dubrawski, A. MOMENT: A family of open time-series foundation models. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.03885.

Ismail Fawaz, H., Lucas, B., Forestier, G., Pelletier, C., Schmidt, D. F., Weber, J., Webb, G. I., Idoumghar, L., Muller, P.-A., and Petitjean, F. InceptionTime: Finding AlexNet for time series classification. *Data Mining and Knowledge Discovery (DMKD)*, 34:1936–1962, 2020.

Lin, T.-Y., Goyal, P., Girshick, R., He, K., and Dollár, P. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)*, pp. 2980–2988, 2017.

Nie, Y., Nguyen, N. H., Seyfi, P., and Kalagnanam, J. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2211.14730.

Ruiz, A. P., Flynn, M., Large, J., Middlehurst, M., and Bagnall, A. The great multivariate time series classification bake off: A review and experimental evaluation of recent algorithmic advances. *Data Mining and Knowledge Discovery (DMKD)*, 35:401–449, 2021.

The PLAsTiCC Team. The photometric LSST astronomical time-series classification challenge (PLAsTiCC), 2018. Zenodo. https://doi.org/10.5281/zenodo.2539456.